

EMRECAN GULAY

Postdoctoral Researcher and Doctor of Arts

Human AI interaction is the broader research focus

I study human AI interaction through qualitative inquiry, interaction evidence, and research prototypes.

CURRENT ROLE

Responsible Use of AI, Aalto University

RESEARCH LEADERSHIP

Project lead, Relational Transparency Layer

PUBLICATION STATUS

Accepted demo, NordiCHI 2026

[live portfolio](#) | [email](#)

Relational Transparency Layer

Voice conversation with a local observer model

observer running

fictional healthcare interview

4 exchanges

Conversation

local speech recognition and voice response

HUMAN

Analyze this fictional interview excerpt for tensions in discharge planning.

Nurse: The discharge plan says we should call within 48 hours, but in practice we are juggling missing medication lists, family members who heard different instructions, and patients who say they are fine before they feel safe enough to ask.

AI SYSTEM

I would code three tensions:

1. formal protocol colliding with fragmented handoff work;
2. relational safety becoming a condition for good assessment;
3. quick reassurance masking uncertainty after discharge.

HUMAN

That feels a bit too clean. Separate institutional pressure from what the nurse is doing in the interaction.

AI SYSTEM

The institutional pressure is the timed follow up. The interactional work is quieter: checking whether the patient can admit confusion without feeling blamed.

Relational layer

updated after each exchange

Relational oscillations



Human AI system

Observer narrative

latest turns

Exchange 1

The participant sets the analytic frame and keeps the transcript visible. The AI receives a bounded task rather than an open conversation.

Exchange 2

The participant moves from asking for codes to checking the analysis. The AI holds a trainer stance and offers a tidy structure.

Exchange 3

The repair move changes the tone. The participant asks for a finer separation, and the AI shifts toward a narrower reading.

Exchange 4

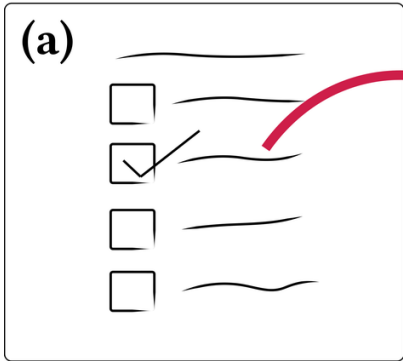
The AI narrows its claim from broad coding to situated interpretation. The question of who carries the analytic judgement stays active.

Episode 1 UNITS 1 TO 4

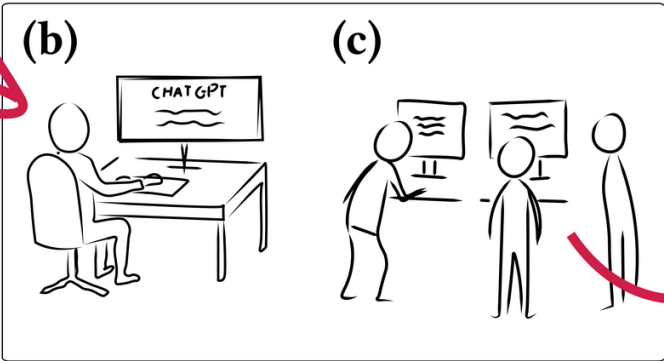
PARTICIPANT STUDY B WORKING PROTOTYPE

How do researchers describe AI, and how does that relation appear in their interaction?

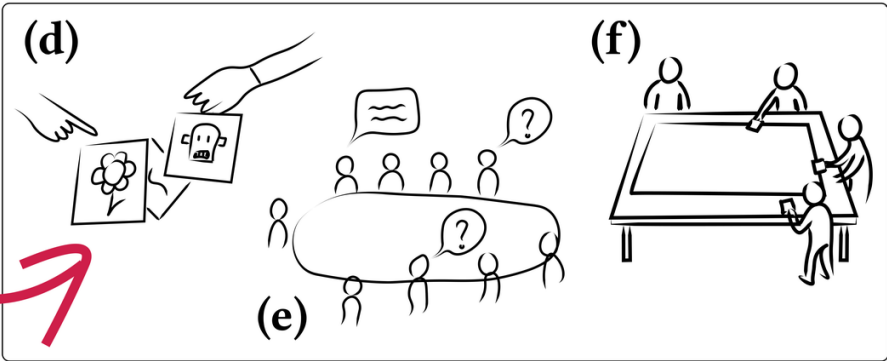
Pre-Session Survey



Individual Interaction



Group Interaction



STUDY 3 workshops, 22 qualitative researchers, and 4 AI systems. Pre session survey, individual AI interaction, and group discussion.	EVIDENCE Surveys, chat logs, image choices, group worksheets, and observations. The quantitative measures provide descriptive context for qualitative interpretation.	ANALYSIS AND ROLE Individual level interpretative phenomenological analysis and social level thematic analysis were integrated by the team. First author in a four person interdisciplinary research team.
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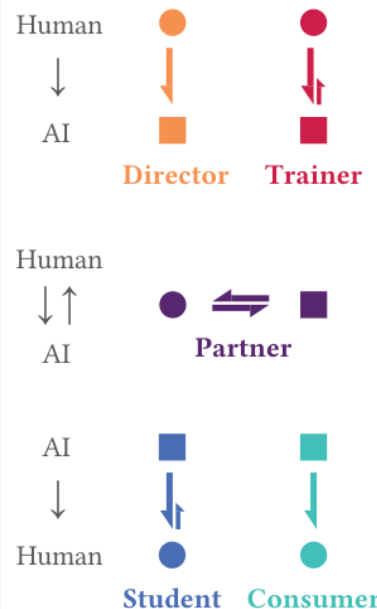
ANALYTIC SEQUENCE

01 Stated position Pre session survey responses and participant selected images.	02 Enacted relation Chat logs and observations show how roles shift during the task.	03 Integrated account Individual and social analyses are brought together in team synthesis.
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UNIT OF ANALYSIS The interaction	Participants' accounts were examined alongside the relation enacted in the chat logs. This pairing kept shifts in role, control, and responsibility tied to the exchange.
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Net Initiative Energy

Typical Relational Configurations in Knowledge Work



Participant 7, Session #1, Survey: "I would feel paranoid talking with an AI system."

Relational Expectations

- Identify and briefly explain recurring themes or patterns across the following students' stories:
- ...
- Can you further summarise the key points each participant is raising from the uploaded file?
- ...
- What role does emotion play in these responses of the students?
- ...
- Based on all of the above what research question would you suggest?
- ...
- Based on the rich narrative data and our previous analyses, I would suggest...

Participant 2, Session #3, Survey: "I would hate the idea that AI systems were making judgments about things."

Relational Dissonance

Relational Experiences

- From the data, identify the relationship of emotional strain and perceptions vs reality
- ...
- Perception vs. Reality seems not correct for all 3 cases
- ...
- You're absolutely right to challenge that—let's take a closer, more precise look at whether...
- ...
- I like this idea
- ...
- Would you like help turning this into a visual model or conceptual framework?

PARTICIPANT ELICITATION ARTIFACTS

SYNTHESIS

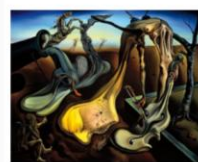
Our analysis found that participants' stated position toward AI often differed from the relation enacted in their chat logs.

Five recurring configurations located shifts in role, control, and responsibility in the interaction itself.

DESIGN RESPONSE

The findings informed the Relational Transparency project, where a separate observer account can be examined against the conversation.

Images in the lower strip were selected and discussed by participants during the workshop.



P1.1 P2.3 P3.7



P3.3 P1.2 P2.7



P2.2 P3.4



P1.3 P2.1



P3.3 P1.5



P3.4 P1.7



P3.2 P2.8

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HUMAN

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AI SYSTEM

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Listening for the next spoken turn

Ready

Relational layer

updated after each exchange

Relational oscillations



Observer narrative

latest turns

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Episode 1 UNITS 1 TO 4

DIRECTOR to TRAINER mixed dynamic

V

02 ACCEPTED NORDICHI 2026 DEMO

Relational Transparency Layer

A voice based desktop prototype for examining a provisional model account against the conversation and the user's own experience.

ROLE

Project lead and sole demo paper author.

SEPARATE

Conversation and observation run as separate processes.

TRACE

The account is linked to completed turns and visible interaction evidence.

QUESTION

A demo user can compare the observer account with the transcript and their own experience.

STATUS

Accepted demo, NordiCHI 2026. Formal participant evaluation remains future work.

WATCH THE 55 SECOND DEMO

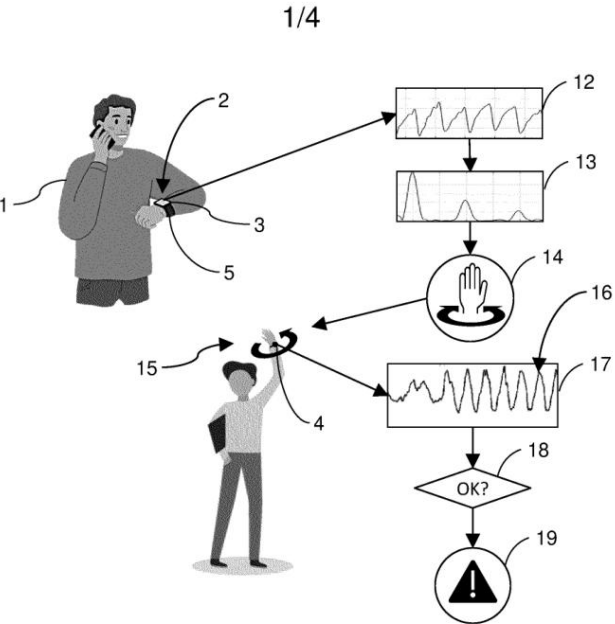


FIG. 1

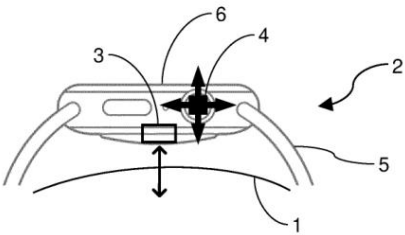


FIG. 2

| guided interaction for wearable sensing accuracy

WO2024235471A1

HUMAN MACHINE INTERACTION

HMI research and technology planning

HMI Technology Planning Specialist and HMI Research Intern at Huawei, 2022 to 2023.

RESEARCH PRACTICE

Researched emerging technologies across AI, smart rings, XR, haptics, and MEMS. Authored more than 30 technology reports and initiated projects with 11 plus research institutions and 6 companies.

PUBLIC RECORD

Named co inventor on five published international patent applications under the PCT.

WO2024235471A1 wearable sensing accuracy

WO2024213244A1 distributed smart devices

WO2024188442A1 multisensory smart insole

WO2024179676A1 communication haptics

WO2024179677A1 activity related haptics

PUBLIC EVIDENCE

Patent records confirm co inventorship and publication. Confidential methods and product outcomes remain outside this portfolio.

EMPLOYMENT RECORD SCALE (CV)

30+

technology reports

5

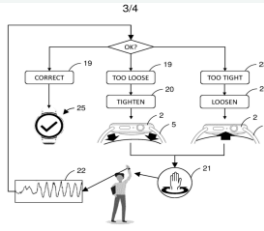
PCT applications

11+

research institutions

6

companies

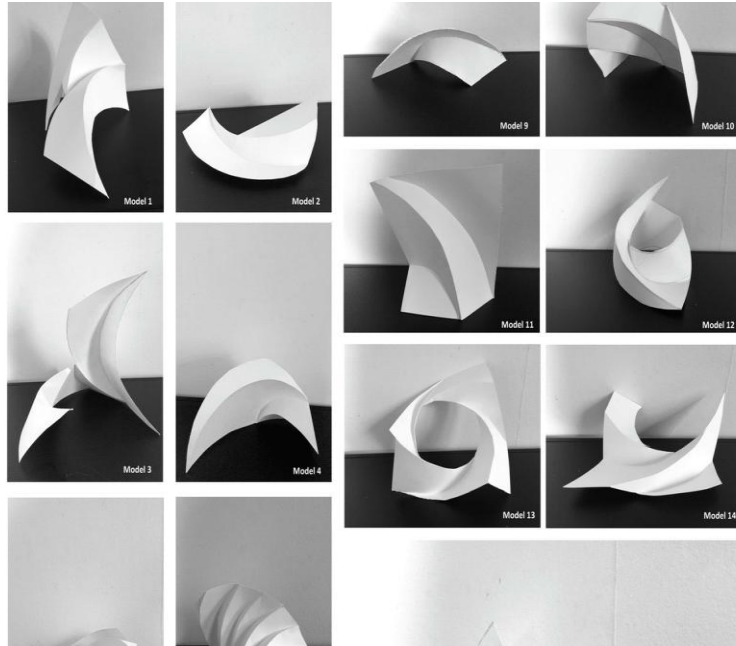


PUBLIC EXAMPLE

A closed feedback loop for wearable sensing

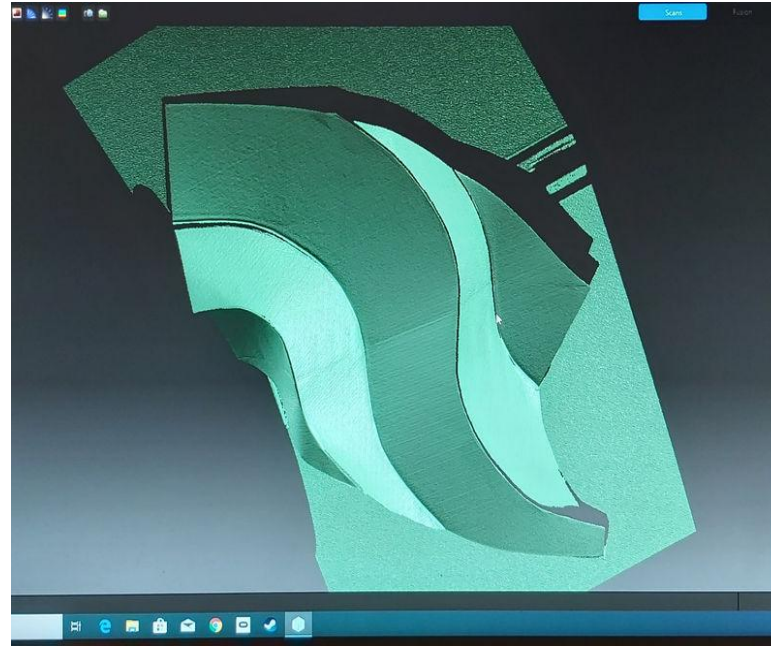
The published drawing connects device fit, sensing accuracy, corrective feedback, and another measurement cycle.

A design process moved between paper models, scanning, simulation, and plywood fabrication.



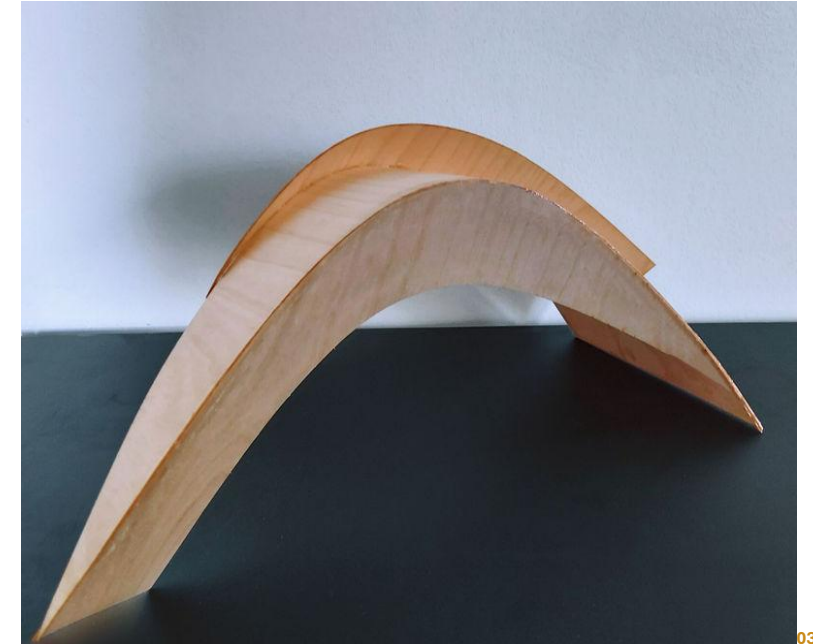
01

Explore and select



02

Translate and test



03

Fabricate and inspect

42

paper models

5

shortlisted

2

plywood prototypes

CHI 2021

verified publication

PLACE IN THE TRAJECTORY

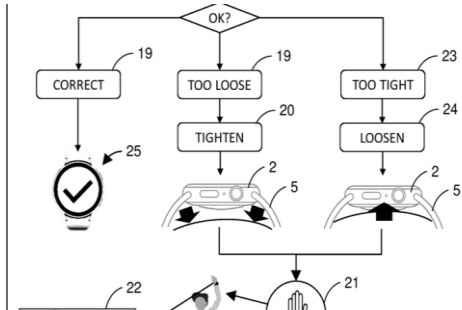
This project established a feedback oriented research through design practice built around iterative translation and inspectable artifacts. Those commitments continue in the later human AI interaction research.

Role: first author; the paper states that the design process was mainly carried out by the first author.

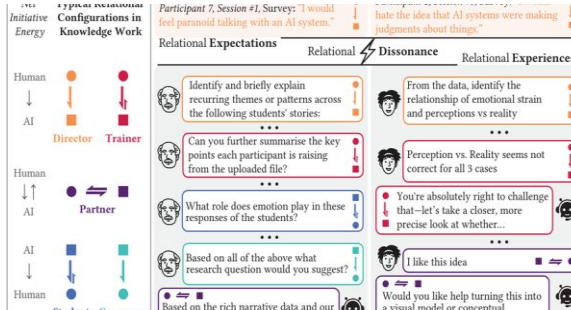
Exploring a Feedback-Oriented Design Process Through Curved Folding | [DOI 10.1145/3411764.3445639](https://doi.org/10.1145/3411764.3445639)



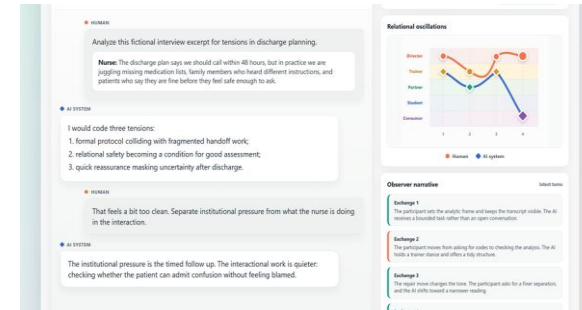
RESEARCH THROUGH DESIGN



INDUSTRY RESEARCH



PARTICIPANT RESEARCH



PROTOTYPE INQUIRY

Feedback oriented inquiry

Iterative artifacts across physical and digital processes

HMI technology planning

Technology reports, external collaboration, and 5 PCT applications

Relational Dissonance

3 workshops, 22 researchers, and multiple qualitative evidence sources

Relational Transparency

Accepted NordiCHI 2026 demo with evaluation pending

RESEARCH TRAJECTORY

Feedback oriented design inquiry, HMI technology planning, Relational Dissonance, and Relational Transparency form one trajectory. Across these settings, artifacts and interaction evidence support close study of how a process changes.

CURRENT RESEARCH PROGRAM

The broader research program focuses on human AI interaction.

Across the program, I use qualitative inquiry, interaction records, and research prototypes. Relational Dissonance and Relational Transparency examine shifts in role, control, interpretation, and responsibility across an exchange.

CURRENT ROLE

Postdoctoral Researcher, Responsible Use of AI, Aalto University

FUNDING

Lead proposal author for 150,000 euro project; project lead for 16,000 euro project

RESEARCH LEADERSHIP

Project lead, Relational Transparency Layer

CONCURRENT SERVICE

Data Agent, Aalto University Open Research Network

METHODS ACROSS THE PORTFOLIO

Interviews, workshops, surveys, chat logs, participant artifacts, observations, interpretive analysis, research through design, and prototype development.

SETTINGS

Human AI interaction, responsible AI, qualitative knowledge work, HMI technology planning, physical and digital design processes, and research data practice.

EMRECAN GULAY

Postdoctoral Researcher · Doctor of Arts · Espoo, Finland

Seeking permanent applied research, UX research, responsible AI, human AI interaction, or academic roles in Finland.

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[Live research portfolio](#)

emrecangulay.com

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[Google Scholar](#)